

ST PETERSBURG, FL, USA

WMO: 997353

Lat: 27.750N

Lon: 82.620W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.8	(c) 8.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 10.2	(k) 18.8	(l) 9.0	(m) 18.5	(n) 3.6	(o) 0	(p) 0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.1	(c) 32.6	(d) N/A	(e) 31.9	(f) N/A	(g) 31.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.4	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 9.6	(b) 8.1	(c) 7.0	(e) 3.4	(f) 34.3	(g) 2.3	(h) 0.8	(i) 1.8	(j) 34.9	(k) 0.5	(l) 35.3	(m) -0.8	(n) 35.8	(o) -2.5	(p) 36.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	16.5	18.4	20.1	23.0	25.8	27.7	28.3	28.8	27.9	25.2	20.5	18.7		
	DBStd	4.96	3.78	3.60	3.05	2.34	1.77	1.31	1.17	1.07	1.24	2.56	3.17	3.79		
	HDD10.0	6	3	1	0	0	0	0	0	0	0	0	0	2		
	HDD18.3	201	78	39	19	3	0	0	0	0	0	1	17	43		
	CDD10.0	4908	206	237	314	390	489	531	567	583	536	470	314	272		
	CDD18.3	2061	22	42	76	143	231	281	308	325	286	213	81	55		
	CDH23.3	18178	23	85	178	696	1797	3012	3527	3924	2997	1608	233	99		
	CDH26.7	5475	0	6	6	79	410	954	1225	1531	921	327	17	1		
(14) Wind	WSAvg	3.2	3.0	3.1	3.6	3.7	3.6	3.5	3.1	3.0	3.0	3.0	2.9	2.8		
(15) Precipitation	PrecAvg	1276	64	68	77	61	64	176	213	219	170	65	39	60		
	PrecMax	1769	139	295	232	219	226	357	385	346	346	167	123	396		
	PrecMin	981	10	11	8	5	6	78	101	118	44	9	6	13		
	PrecStd	203	39	59	60	43	54	76	66	70	80	35	29	96		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.4	27.7	27.5	29.6	32.5	33.3	33.1	33.7	33.0	31.6	28.6	26.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	24.1	25.6	26.2	28.4	31.0	31.9	32.1	32.6	31.9	30.3	26.9	25.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	22.8	24.2	25.2	27.7	29.9	31.1	31.4	31.8	31.1	29.4	25.8	24.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	DB	21.6	23.1	24.3	26.7	28.9	30.3	30.6	31.2	30.4	28.5	24.9	23.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	7.1	6.8	6.5	6.2	5.9	5.3	5.0	5.1	5.4	5.6	6.5	6.4		
		MCWBR	7.0	7.0	6.9	7.0	7.2	6.2	5.8	5.7	6.0	5.9	6.4	6.3		
(40) Clear-Sky Solar Irradiance	taub	taud	0.343	0.356	0.369	0.393	0.423	0.447	0.471	0.451	0.422	0.390	0.355	0.344		
		Ebn at Noon	2.529	2.499	2.448	2.341	2.316	2.330	2.269	2.337	2.383	2.461	2.505	2.550		
	Edh at Noon		904	918	923	904	872	846	824	841	858	870	882	886		
			90	100	111	128	131	129	136	127	118	103	92	85		
(44) All-Sky Solar Radiation	RadAvg		3.56	4.31	5.44	6.53	6.92	6.27	5.94	5.71	5.18	4.79	3.90	3.30		
	RadStd		0.26	0.28	0.45	0.31	0.37	0.51	0.36	0.35	0.34	0.29	0.20	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.43	+0.80	N/A	N/A	N/A	N/A	-3	-45	+173	+121

Nomenclature: See separate page